

Number-Theoretic Transform in Cryptography

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1 Introduction

The purpose of this note is to derive the Number-Theoretic Transform (NTT) for general partially splitting primes. Explicit formulas are given in Section 3, while a conceptual exposition of the construction appears in Section 2. This exposition may be of independent interest because the same theory of prime ideal factorization underpins nearly all the relevant algebraic structures in lattice-based cryptographic primitives.

Parameter instantiation is a challenge in lattice-based cryptography, as algebraic structures depend on choices such as the prime modulus and the base ring. Such dependencies can be subtle for researchers and practitioners unfamiliar with algebraic number theory. In practice, subtle bugs often arise. For example, choosing a completely split prime at an insufficient ring dimension in [BC26] compromises knowledge-extraction soundness.

Beyond security, performance optimization is challenging, especially because performance and security share dependencies on the base prime. For instance, conventional NTT constructions require a prime that splits completely in the ring of integers. Such primes have residue degree 1, which enables fast NTT operations but diminishes the challenge space needed for knowledge-extraction soundness (see [LS18] for an explicit formula for the size of the challenge space). Moreover, optimizing lattice arithmetic is highly nonmodular. For example, the choice of prime affects both the Hamming weight and the residue degree in highly nontrivial ways that reflect number-theoretic properties. Finding suitable primes is therefore challenging but offers a promising optimization opportunity [NS26].

We hope that the exposition serves the following purposes:

- Give a conceptual exposition of the construction of the NTT for general partially splitting cases.
- Make clear the dependence on parameters.

2 Background

Let K be a finite extension of the field of rational numbers $\mathbb{Q}$. Use $\mathcal{O}_K$ to denote the ring of integers of K .

Prime Factorization in Number Fields To understand the NTT construction and, more generally, the algebraic structures underlying lattice-based cryptography constructed from algebraic lattices, one needs to understand prime factorization in the ring of integers of a number field. Recall that a prime ideal $\mathfrak{p} \subseteq \mathcal{O}_K$ is a set closed under addition and scalar multiplication by elements of $\mathcal{O}_K$ such that, if $ab \in \mathfrak{p}$, then $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Prime factorization in the naive sense fails for algebraic integers. For example, in $\mathbb{Z}[i]$, the ring of Gaussian integers, where i is the imaginary unit satisfying $i^2 = -1$, the number 5 is not prime because $5 = (2 + i)(2 - i)$. However, there is a notion of prime factorization for ideals in rings of integers. In particular, any nonzero ideal I can be expressed as a product of ideals $I = \prod_{i=1}^m \mathfrak{p}_i^{e_i}$, where the $\mathfrak{p}_i$ are prime ideals and the e_i are positive integers. Moreover, because rings of integers are Dedekind domains, every such ideal $\mathfrak{p}_i$ is maximal, meaning that $\mathcal{O}_K/\mathfrak{p}_i$ is a field. The map $\mathcal{O}_K \rightarrow \mathcal{O}_K/\mathfrak{p}_i$ is the “restriction” map to the residue field $\mathcal{O}_K/\mathfrak{p}_i$, which allows one to study the algebraic structures one prime at a time. Dedekind and Kummer developed this perspective in the middle of the 19th century. They observed remarkable similarities between algebraic number theory and the geometry of Riemann surfaces, inspiring the development of classical algebraic geometry.

Unramified Primes By a rational prime number, we mean an ordinary prime number in the ring of integers. A rational prime number p is called “unramified” if all the prime factors in the prime ideal factorization of (p) have exponent 1, meaning that $(p) = \prod_{i=1}^m \mathfrak{p}_i$ for distinct prime ideals $\mathfrak{p}_i$. The name “ramification” suggests that this notion can be understood from a geometric perspective independent of number theory, but we do not discuss that perspective because it is not required for cryptography. In practice, a prime is unramified if and only if it does not divide the absolute discriminant of K , which can be computed readily. For a cyclotomic extension, every prime that does not divide the conductor is unramified. Specifically, for a cyclotomic extension whose conductor is a power of 2, a prime p is unramified if and only if it is odd. For the rest of this note, we assume that the prime p is unramified. Readers curious about why the neurons in the visual cortex of arithmetic geometers light up can refer to Appendix A.

NTT For an unramified prime p , the Chinese Remainder Theorem states that

$$\mathcal{O}_K/p\mathcal{O}_K \cong \prod_{i=1}^m \mathcal{O}_K/\mathfrak{p}_i \quad (1)$$

The NTT map is simply the map given by

$$\text{NTT} : x \bmod p \mapsto (x \bmod \mathfrak{p}_1, \dots, x \bmod \mathfrak{p}_m).$$

Algebraic geometers recognize this as simply the functor of “evaluating” on all stalks, which concretely means all information about $\mathcal{O}_K/p\mathcal{O}_K$ can be recovered from local information at each prime $\mathfrak{p}_i$.

In the proof of the Chinese Remainder Theorem, one constructs so-called “idempotents”¹ e_i satisfying the relations $e_i^2 = e_i$, $e_i e_j = 0$ for $i \neq j$, and $\sum_{i=1}^m e_i = 1$. The inverse NTT map

¹The idempotents correspond geometrically to the connected components. This makes the interpretation intuitive but of course is not needed for cryptography.

is then given by

$$\text{INTT} : (x_1, \dots, x_m) \mapsto x_1 e_1 + \dots + x_m e_m.$$

3 Specialization to $\mathbb{Z}[\zeta] = \mathbb{Z}[x]/(x^n + 1)$

Assume that n is a power of 2. Let ζ be a primitive $2n$ -th root of unity and consider the cyclotomic field $K = \mathbb{Q}(\zeta)$. Its ring of integers $\mathcal{O}_K$ is isomorphic to $\mathbb{Z}[\zeta] = \mathbb{Z}[x]/(x^n + 1)$. We specialize the construction in Section 2 to this case, which is common in cryptography. We first present a formula for the partially splitting case, which generalizes to the fully splitting case common in many cryptographic applications. In practice, one often chooses a partially splitting prime instead of a fully splitting one for knowledge-extraction soundness reasons.

NTT formula for the partially splitting primes Since the extension $K/\mathbb{Q}$ is Galois, the polynomial $x^n + 1$, which is the minimal polynomial of ζ over $\mathbb{Q}$, splits into m distinct irreducible polynomial factors of equal degree d over $\mathbb{F}_p$ for an odd prime p ; that is,

$$x^n + 1 = \prod_{i=1}^m f_i(x)$$

where the $f_i(x)$ are irreducible polynomials of degree d over $\mathbb{F}_p$. We thus have

$$\mathbb{Z}[x]/(p, f_i)\mathbb{Z}[x] \cong \mathbb{F}_p[x]/(f_i(x)) \cong \mathbb{F}_p[\alpha_i] \cong \mathbb{F}_{p^d},$$

where α_i is a root of $f_i(x)$ and d is the residue degree, which, by standard Galois theory, is equal to $\text{ord}_{2n}(p)$; that is, d is the smallest positive integer such that $p^d \equiv 1 \pmod{2n}$. Hence, the isomorphism in the Chinese Remainder Theorem is given by

$$\begin{aligned} \text{NTT} : \mathbb{F}_p[x]/(x^n + 1) &\longrightarrow \prod_{i=1}^m \mathbb{F}_{p^d}, \\ f(x) \bmod (x^n + 1) &\longmapsto (f(\alpha_1), \dots, f(\alpha_m)). \end{aligned} \tag{2}$$

Remark 1. One can interpret the NTT formula (2) as the evaluation of the polynomial $f(x)$ at the roots $\alpha_1, \dots, \alpha_m$ of the polynomial $x^n + 1$. Hence the NTT is a version of the familiar Reed-Solomon code.

Inverse NTT formula for partially splitting primes From Remark 1, one can derive the inverse NTT formula from standard Lagrange interpolation techniques on evaluation points of polynomials. We give a derivation for completeness.

Fix a finite field F over which $x^n + 1$ splits completely; that is,

$$x^n + 1 = \prod_{\beta \in \Omega} (x - \beta),$$

where $\Omega = \{\beta \in F \mid \beta^n + 1 = 0\}$. The Lagrange interpolation formula gives

$$f(x) = \sum_{\beta \in \Omega} f(\beta) \prod_{\beta' \in \Omega \setminus \{\beta\}} \frac{x - \beta'}{\beta - \beta'}.$$

Expanding this expression gives

$$\begin{aligned} f(x) &= \sum_{\beta \in \Omega} f(\beta) \left(\frac{1}{n} \sum_{j=0}^{n-1} \beta^{-j} x^j \right) \\ &= \sum_{j=0}^{n-1} \left(\frac{1}{n} \sum_{\beta \in \Omega} f(\beta) \beta^{-j} \right) x^j. \end{aligned}$$

Using standard techniques that express a sum of Galois conjugates as a trace by counting Frobenius orbits, one can show that

$$\frac{1}{n} \sum_{\beta \in \Omega} f(\beta) \beta^{-j} = \frac{1}{n} \sum_{i=1}^m \text{Tr}_{\mathbb{F}_{p^d}/\mathbb{F}_p} (f(\alpha_i) \alpha_i^{-j}), \quad (3)$$

where $\alpha_1, \dots, \alpha_m$ are the chosen roots described in (2). As a result, we obtain the inverse NTT formula

$$\begin{aligned} \text{INTT}: \prod_{i=1}^m \mathbb{F}_{p^d} &\longrightarrow \mathbb{F}_p[x]/(x^n + 1), \\ (a_1, \dots, a_m) &\longmapsto \frac{1}{n} \sum_{j=0}^{n-1} \left(\sum_{i=1}^m \text{Tr}_{\mathbb{F}_{p^d}/\mathbb{F}_p} (a_i \alpha_i^{-j}) \right) x^j. \end{aligned} \quad (4)$$

Appendix A Ramification

Ramification has a natural geometric interpretation. A ring homomorphism $A \rightarrow B$ induces a map $\text{Spec } B \rightarrow \text{Spec } A$ that sends $\mathfrak{q}$ to $\mathfrak{q} \cap A$, where $\text{Spec } A$ is the set of prime ideals of A .

In the cryptographic setting, take $A = \mathbb{Z}$ and $B = \mathbb{Z}[\zeta]$, where ζ is a primitive $2n$ -th root of unity. A rational prime factors as $pB = \prod_{i=1}^m \mathfrak{p}_i^{e_i}$. Here e_i is the ramification index and $f_i = [B/\mathfrak{p}_i : \mathbb{F}_p]$ is the residue degree. They satisfy $\sum_{i=1}^m e_i f_i = [\mathbb{Q}(\zeta) : \mathbb{Q}] = n$. Because the extension is Galois, all the e_i are equal, as are all the f_i . The prime p is unramified precisely when every $e_i = 1$. Geometrically, $\text{Spec } B \rightarrow \text{Spec } \mathbb{Z}$ is then finite étale over (p) and locally behaves like a finite covering map. This viewpoint extends ramification beyond number fields to more general geometric settings.

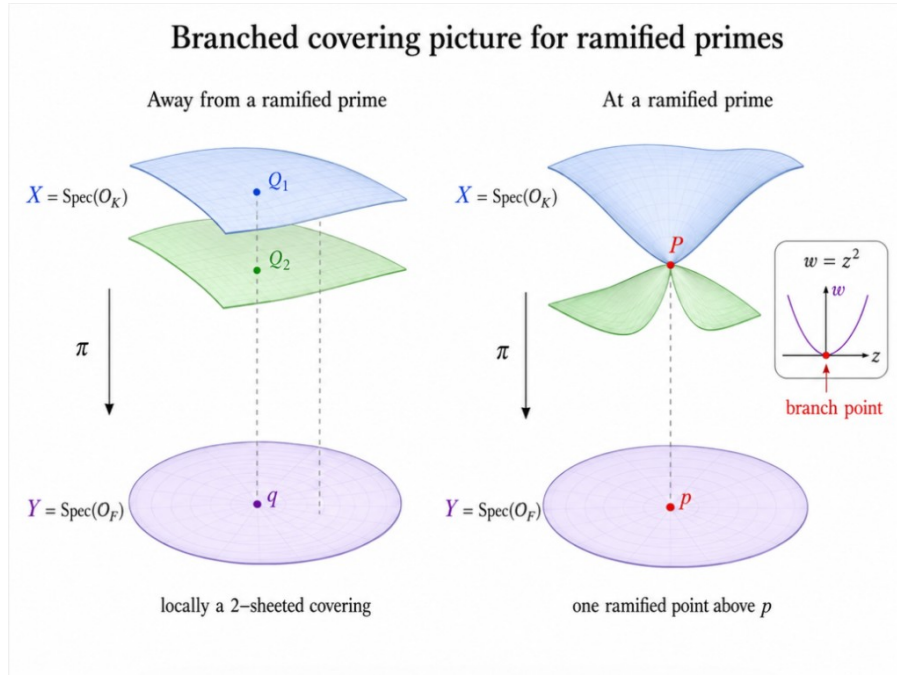


Figure 1: The geometric interpretation of ramification as branching.

References

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